

1 Highlights

2 **Interpolation versus generalization in machine learning prediction** 3 **of micropollutant adsorption on layered double hydroxides**

4 [AUTHORS TO BE ADDED]

- 5 • 1,342 LDH adsorption records from 59 studies evaluated leakage-aware
- 6 • Random splits overstate ML skill: R^2 0.85 within studies, near zero
7 across
- 8 • Leave-one-study-out audit quantifies transfer for every source study
- 9 • Conformal intervals certify interpolation; new-study intervals widen 3–
10 $4\times$
- 11 • $\text{pH}-\text{pH}_{\text{pzc}}$ feature links model behavior to electrostatics

12 Interpolation versus generalization in machine learning
13 prediction of micropollutant adsorption on layered
14 double hydroxides

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16 **Abstract**

17 Micropollutants such as pharmaceuticals, pesticides, and personal-care prod-
18 ucts are detected in water bodies worldwide, and layered double hydroxides
19 (LDHs) are widely studied as tunable adsorbents for their removal. Machine
20 learning (ML) models trained on literature-compiled datasets increasingly
21 report near-perfect prediction of adsorption performance, yet the evaluation
22 protocols behind these reports rarely account for the structure of the under-
23 lying data, in which many experimental records originate from few source
24 studies. We compiled 1,342 adsorption experiments for LDH–micropollutant
25 systems from 59 published studies and evaluated seven regression models (lin-
26 ear regression, k -nearest neighbors, neural network, Cubist, random forest,
27 XGBoost, and histogram gradient boosting) under a leakage-aware protocol
28 that separates two questions: interpolation within studied systems and gen-
29 eralization to unseen studies. With duplicate records removed and all prepro-
30 cessing fitted on training folds only, the best models predicted removal ratio
31 (RR%) for studied systems at new operating conditions with R^2 of 0.85 ± 0.03
32 and RMSE near 11 percentage points across 20 repeated splits. Under leave-
33 one-study-out evaluation, however, the median per-study error rose to 14–16

percentage points, two to three times the interpolation error, and only 20 of the 59 studies were predicted with accuracy resembling the interpolation regime, showing that random-split protocols overstate the screening ability of literature-trained models by a wide margin; the models still reduce the error of descriptor-free baselines by roughly 40 percent, so some signal does transfer. Split-conformal prediction intervals certified the interpolation task (93% empirical coverage at a 90% nominal level, with ± 19 -point intervals), but for unseen studies the intervals had to widen to ± 68 points to maintain coverage, and a common distance-to-training applicability score carried no information about realized error in either task. Chemistry-informed features, including the electrostatic driving force $\text{pH} - \text{pH}_{\text{pzc}}$, changed the errors little yet yielded feature attributions consistent with electrostatic and hydrophobic adsorption mechanisms. The results delimit what literature-trained adsorption models can currently support in practice: condition optimization for studied systems, but not yet the screening of new adsorbent–pollutant combinations.

Keywords: layered double hydroxide, micropollutant, adsorption, machine learning, data leakage, uncertainty quantification

52 Nomenclature

LDH	layered double hydroxide
RR%	removal ratio (percent)
C_i	initial concentration (mg/L)
AD	adsorbent dose (g/L)
pH _{pzc}	pH at the point of zero charge
BET	Brunauer–Emmett–Teller surface area (m ² /g)
PV	pore volume (cm ³ /g)
APD	average pore diameter (nm)
d_{003}	basal interlayer spacing (nm)
53 log K_{ow}	octanol–water partition coefficient
ML	machine learning
LR, KNN, ANN	linear regression, k -nearest neighbors, neural network
RF, XGB, HistGB	random forest, XGBoost, histogram gradient boosting
LOSO	leave-one-study-out
(T1), (T2)	interpolation task, cross-study generalization task
MAE, RMSE	mean absolute error, root mean square error
SHAP	Shapley additive explanations
CV	cross-validation

54 1. Introduction

55 Micropollutants, including pharmaceuticals, pesticides, endocrine disruptors,
56 and personal-care products, are detected at ng/L to µg/L levels in
57 surface water, groundwater, and treated effluent worldwide (TODO, 0000d).
58 Conventional treatment trains remove many of these compounds only par-

59 tially, and their chronic ecological and human-health effects have placed
60 micropollutant control on regulatory agendas in the European Union, the
61 United States, and Asia (TODO, 0000e). Adsorption is among the most
62 widely deployed polishing steps because it requires no reagent input, pro-
63 duces no transformation by-products, and operates across wide concentration
64 ranges (TODO, 0000b).

65 Layered double hydroxides (LDHs) are a class of anionic clays with the
66 general formula $[M_{1-x}^{2+}M_x^{3+}(\text{OH})_2]^{x+}[A^{n-}]_{x/n} \cdot m\text{H}_2\text{O}$, whose metal composi-
67 tion, interlayer anion, and calcination state can be tuned to target specific
68 pollutants (TODO, 0000c). Hundreds of LDH variants have been synthesized
69 and tested against individual micropollutants, and each such study reports
70 removal under its own grid of pH, temperature, contact time, adsorbent dose,
71 and initial concentration. The resulting literature is large but fragmented: no
72 single laboratory can cover the combinatorial space of metal pairs, synthesis
73 routes, and pollutants experimentally.

74 Isotherm and kinetic models describe each of these systems in isolation
75 and offer no mechanism for pooling evidence across studies. Machine learning
76 (ML) offers a natural response to this fragmentation. By compiling published
77 experiments into a single dataset, ML models have been trained to predict
78 adsorption capacity or removal ratio from material, pollutant, and operat-
79 ing descriptors, with reported test R^2 values frequently above 0.9 (TODO,
80 0000f,g,a). Such models promise two distinct services: interpolating the be-
81 havior of an already-studied adsorbent–pollutant system at untested condi-
82 tions, and screening new systems before synthesis. The second service is the
83 one most frequently invoked to motivate these studies.

84 However, we argue that the evaluation protocols behind these reports do
85 not support the screening claim. Firstly, literature-compiled datasets are
86 strongly clustered: in our compilation, 1,342 experimental records originate
87 from only 59 source studies, and within one study the material and pollutant
88 descriptors are constant while only operating conditions vary. A random
89 record-level train–test split therefore places near-duplicate records on both
90 sides, and the test score measures within-study interpolation while being
91 reported as generalization. Secondly, preprocessing steps such as imputa-
92 tion, scaling, and feature selection are commonly fitted on the full dataset
93 before splitting, which leaks test-set information into training (Kapoor and
94 Narayanan, 2023). Thirdly, point predictions are reported without uncer-
95 tainty estimates or applicability bounds, leaving practitioners no way to know
96 when a prediction should be trusted.

97 To address the above challenges, we evaluate ML models for micropollu-
98 tant adsorption on LDHs under a protocol that closes each of these leakage
99 pathways. In essence, our idea is to treat the source study as the unit of
100 generalization: we evaluate every model twice, once under a duplicate-free
101 record-level split that answers the interpolation question, and once under
102 grouped, leave-one-study-out protocols that answer the screening question,
103 with all preprocessing confined to training folds in both cases. We comple-
104 ment point predictions with split-conformal intervals and a distance-based
105 applicability-domain score, and we examine whether chemistry-informed fea-
106 tures improve cross-study transfer.

107 More specifically, this study aims to (1) quantify the gap between inter-
108 polation and cross-study generalization for seven regression models on 1,342

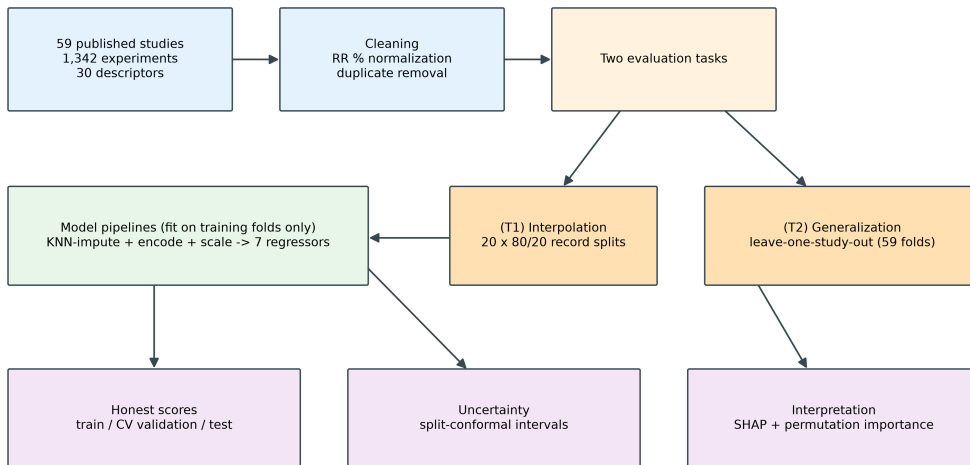


Figure 1: Evaluation workflow

LDH-micropollutant records from 59 studies; (2) equip the predictions with distribution-free uncertainty intervals and an applicability-domain flag, and test their reliability under both tasks; and (3) evaluate chemistry-informed features, including the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, and read the resulting feature attributions against known adsorption mechanisms. To the best of our knowledge, this is the first study of LDH adsorption modeling that separates these two prediction tasks explicitly and reports what literature-trained models can, and cannot yet, deliver for practice.

2. Materials and methods

Figure 1 summarizes the workflow: compilation and cleaning of the literature data, the two-task evaluation protocol, leakage-controlled model pipelines, and the uncertainty and interpretation layers.

121 2.1. Data curation

122 We compiled 1,342 batch-adsorption experiments for LDH–micropollutant
 123 systems from 59 peer-reviewed studies. Each record holds 30 descriptors in
 124 four groups: micropollutant properties (compound identity, $\log K_{ow}$, molecu-
 125 lar weight, the Abraham solvation parameters E , S , A , B , V , and topological
 126 polar surface area), LDH composition and synthesis (divalent and trivalent
 127 metal, molar ratio, synthesis route, hydrothermal temperature and time, cal-
 128 cination conditions, and composition-averaged elemental descriptors), mate-
 129 rial texture (interlayer spacing d_{003} , BET surface area, pore volume, average
 130 pore diameter, and point of zero charge pH_{pzc}), and operating conditions
 131 (pH, temperature, contact time, adsorbent dose, and initial concentration).
 132 Table 1 summarizes the numeric descriptors; the prediction target is the
 133 removal ratio (RR%).

134 **Cleaning.** Values stored as fractions were converted to percent by a
 135 range-aware rule that leaves already-percent data untouched, and category
 136 labels were normalized (e.g., merging spelling variants of *co-precipitation*).
 137 Records identical in every descriptor were reduced to a single occurrence,
 138 since duplicates spanning a train–test boundary would let the test be an-
 139 swered by lookup; deduplication removed 83 records. Missing texture val-
 140 ues were imputed by a k -nearest-neighbor imputer computed in standardized
 141 space and mapped back to physical units; the imputer, like every fitted trans-
 142 form in this work, was trained on training folds only. The source study of
 143 each record is retained as a grouping key and is never used as a model input.
 144 Details of the cleaning rules are in Text S1.

145 *2.2. Evaluation protocol*

146 **Two prediction tasks.** We evaluate every model under two protocols
147 that correspond to the two services expected from literature-trained models.
148 Task (T1), *within-system interpolation*, asks how well a model predicts RR%
149 for a studied adsorbent–pollutant system at new operating conditions. Task
150 (T2), *cross-study generalization*, asks how well the model predicts systems it
151 has never seen. Intuitively, (T1) allows the model to have seen the material
152 and the pollutant at other conditions, while (T2) does not.

153 **Repeated splits for (T1).** A single random split can be lucky or un-
154 lucky, so (T1) is measured over 20 seeded 80/20 record-level splits of the
155 duplicate-free data, and scores are reported as mean $\pm$ standard deviation
156 across repeats. Hyperparameters are tuned once (Section 2.3) and held fixed
157 across repeats, so the spread reflects sampling variability rather than tuning
158 noise.

159 **Leave-one-study-out audit for (T2).** Each of the 59 studies s is
160 held out once; the model f_{-s} is refit on the remaining 58 studies with fixed
161 hyperparameters and scored on the held-out records:

$$\text{MAE}_s = \frac{1}{n_s} \sum_{i \in s} |y_i - f_{-s}(\mathbf{x}_i)|. \quad (1)$$

162 Reporting the distribution of MAE_s across studies, rather than one pooled
163 score, exposes which systems transfer and which do not. A learning curve
164 over the number of training studies tests whether cross-study error is limited
165 by data volume or by data diversity.

166 **Naive baselines.** To make the error magnitudes interpretable, we score
167 three predictors that use no descriptors: the global training-mean; the per-
168 pollutant mean (the average RR% of the same compound in the training

169 data, falling back to the global mean for unseen compounds); and, for (T1)
 170 only, the per-study mean. A model is useful for screening only if it beats
 171 these lookups on unseen studies.

172 **Leakage control.** Imputation, scaling, and categorical encoding are
 173 embedded in each model pipeline and fitted on training folds only. Hyperpa-
 174 rameter tuning uses cross-validation folds that differ from the folds used for
 175 reporting, and grouped tasks use group-aware folds throughout. These con-
 176 trols address the leakage mechanisms documented for applied ML (Kapoor
 177 and Narayanan, 2023), which we found to inflate reported skill substantially
 178 when absent (Section 3.3).

179 2.3. Regression models

180 We compare seven regressors spanning the model families used in the
 181 adsorption-ML literature, chosen so that their inductive biases differ. Through-
 182 out, $\mathbf{x}_i \in \mathbb{R}^p$ denotes the preprocessed descriptor vector of record i and $\hat{y}_i$ its
 183 predicted RR%.

184 **Linear regression (LR)** fits one additive coefficient per descriptor,

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j, \quad (2)$$

185 with β estimated by least squares. It serves as the interpretable floor: any
 186 nonlinearity or interaction in the adsorption response appears as a gap be-
 187 tween LR and the flexible models.

188 **k -nearest neighbors (KNN)** predicts the average target of the k closest
 189 training records in standardized feature space (Cover and Hart, 1967),

$$\hat{y}(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x})} y_i, \quad (3)$$

190 where $\mathcal{N}_k(\mathbf{x})$ holds the k nearest neighbors under Euclidean distance and
 191 k is grid-searched between 1 and 9. KNN has no functional form at all;
 192 its performance measures how far pure similarity lookup carries prediction,
 193 which makes it a diagnostic for near-duplicate structure in the data.

194 **Artificial neural network (ANN)** is a single-hidden-layer perceptron
 195 trained by Adam (Rumelhart et al., 1986),

$$\hat{y} = \mathbf{w}^\top \sigma(W\mathbf{x} + \mathbf{b}) + b_0, \quad (4)$$

196 with 32 hidden units and rectified-linear activation $\sigma(z) = \max(0, z)$, match-
 197 ing the compact architectures used in prior adsorption studies. Larger archi-
 198 tectures were not explored because the dataset holds fewer than 10^3 effective
 199 degrees of freedom per feature.

200 **Cubist** builds a set of rules R_m whose leaves hold linear models (Quinlan,
 201 1992),

$$\hat{y}(\mathbf{x}) = \text{avg}_{m: \mathbf{x} \in R_m} \left(\beta_0^{(m)} + \sum_j \beta_j^{(m)} x_j \right), \quad (5)$$

202 optionally smoothed toward the targets of nearby training instances; commit-
 203 tees and neighbors are grid-searched. Unlike constant-leaf trees, its leaf-linear
 204 structure can extrapolate moderately beyond the training range of a rule, a
 205 property that becomes relevant under (T2).

206 **Random forest (RF)** averages B decorrelated trees grown on bootstrap
 207 samples (Breiman, 2001),

$$\hat{y}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}), \quad (6)$$

208 providing a variance-reduced piecewise-constant baseline among the ensem-
 209 bles.

210 **Gradient-boosted trees** build the prediction stagewise, each new tree
 211 fitting the residual structure left by its predecessors,

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x}), \quad (7)$$

212 with learning rate ν and base trees h_m . **XGBoost** implements Eq. (7) with
 213 second-order updates and a regularized objective that penalizes the number
 214 of leaves and their weights (Chen and Guestrin, 2016); **histogram gradi-**
 215 **ent boosting (HistGB)** is the scikit-learn implementation with histogram-
 216 binned split finding (Ke et al., 2017; Pedregosa et al., 2011). Both are the
 217 strongest current defaults for tabular data.

218 Tree-ensemble hyperparameters were tuned with the TPE sampler of Op-
 219 tuna over 30 trials (Akiba et al., 2019) on cross-validation folds of the training
 220 data. Scale-sensitive models (LR, KNN, ANN, Cubist) receive a standardizer
 221 inside their pipeline; tree ensembles operate on raw feature scales. Search
 222 spaces and selected values are in Table S1; all splits, folds, and samplers are
 223 seeded for reproducibility.

224 2.4. Feature screening and selection

225 Multicollinearity is diagnosed with the variance inflation factor,

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (8)$$

226 where R_j^2 is the coefficient of determination of descriptor j regressed on all
 227 other descriptors (targets excluded). Because descriptors are constant within
 228 a study, the study-cluster structure drives VIF values orders of magnitude
 229 beyond the usual threshold of 10 (Section 3.1), so VIF is used here as a
 230 redundancy diagnostic rather than an automatic elimination rule.

231 Feature selection is then evaluated the same way as the models them-
 232 selves: three nested descriptor sets — the full set of 30, a correlation-screened
 233 set of 21 that drops the collinear Abraham and composition-averaged block,
 234 and the 10 descriptors with the best median permutation-importance rank
 235 — are compared under both tasks with fixed hyperparameters. Judging a
 236 feature set only under (T1) would inherit the same optimism as judging a
 237 model there, so both tasks are reported.

238 2.5. Uncertainty and applicability domain

239 **Split-conformal intervals.** We wrap the best model in split-conformal
 240 prediction (Vovk et al., 2005; Lei et al., 2018). A calibration subset of n
 241 records is withheld from training, absolute residuals $r_i = |y_i - \hat{y}_i|$ are com-
 242 puted on it, and the interval half-width is their conformal quantile

$$q = \text{Quantile}\left(\{r_i\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right), \quad (9)$$

243 so that each test prediction is reported as $[\hat{y} - q, \hat{y} + q]$ with nominal coverage
 244 $1 - \alpha = 0.9$. For task (T2) the calibration subset is drawn as whole studies,
 245 so the calibration residuals reflect the new-study setting. Conformal validity
 246 assumes exchangeability between calibration and test units, which grouped
 247 data satisfy only approximately; we therefore report empirical coverage for
 248 both tasks rather than assuming it.

249 **Applicability domain.** As a candidate trust flag we compute, for each
 250 test record, the mean standardized distance to its five nearest training records
 251 in the preprocessed feature space. Records far from the training distribution
 252 should receive predictions with caution; we validate the flag by correlating it
 253 with realized absolute error rather than assuming it works.

254 2.6. Feature engineering and attribution

255 Beyond the raw descriptors, we construct three features with mechanistic
 256 meaning: the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, whose sign determines
 257 whether the LDH surface is positively or negatively charged relative to the
 258 solution; the initial loading ratio C_i/dose ; and the interaction $\log K_{\text{ow}} \times$
 259 $(\text{pH}-\text{pH}_{\text{pzc}})$ coupling hydrophobicity with surface charge. Both tasks are
 260 re-evaluated with and without these features. Feature attributions are com-
 261 puted with SHAP (Lundberg and Lee, 2017) on the held-out test set using
 262 the fitted models, complemented by permutation importance ranks across all
 263 seven models and one-dimensional partial dependence profiles, and the re-
 264 sulting patterns are read against the expected electrostatic and hydrophobic
 265 mechanisms.

266 2.7. Performance metrics

267 Model skill is reported as the coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (10)$$

268 the root mean square error, $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$, and the mean abso-
 269 lute error, $\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$, each computed on three bases with distinct
 270 meanings: the training records (in-sample fit), cross-validation folds of the
 271 training set (validation), and the held-out test records of the task at hand.
 272 All three are reported to keep the in-sample ceiling and the generalization
 273 estimate visibly separate. Wall-clock training and prediction times are re-
 274 ported in Section 3.7.

275 3. Results and discussion

276 Section 3.1 first characterizes the compiled dataset. The modeling ex-
277 periments then answer four research questions. (RQ1) How accurately do
278 the models predict RR% for studied systems at new operating conditions?
279 (RQ2) Does this accuracy transfer to systems from unseen studies? (RQ3) Can
280 the uncertainty of individual predictions be quantified reliably under both
281 tasks? (RQ4) Which factors govern the predictions, and are they chemically
282 plausible?

283 3.1. Dataset characteristics

284 Figure 2 summarizes the compiled data. The target is broad and near-
285 symmetric (RR% mean 45.5, median 43.7, SD 28.0, skewness 0.11) and spans
286 the full 0–100% range, but its distribution differs strongly between pollutants
287 and between LDH metal pairs (Fig. 2a,b), reflecting both genuine chemistry
288 and the uneven attention different systems have received in the literature.
289 The compilation covers 14 micropollutants, 13 metal pairs, and 4 synthe-
290 sis routes, with record and study counts diverging sharply: diclofenac con-
291 tributes 304 records from only 3 studies, while ciprofloxacin contributes 181
292 records spread over 32 studies. The study-size distribution is likewise skewed:
293 the largest single study contributes 220 of the 1,342 records, the median study
294 only 17, and a random record-level sample is therefore dominated by a hand-
295 ful of intensively measured systems. Several operating variables have heavy
296 right tails (skewness 5–6 for dose, initial concentration, and contact time;
297 Table S4), which the rank-based correlation analysis below accommodates.

298 **Univariate association with the target.** Table 2 lists the Spearman

correlation of each numeric descriptor with RR%. No descriptor exceeds $|\rho| = 0.43$ (pore volume), and the leading operating variables reach only 0.27–0.33, so no single factor controls removal monotonically. This is because the response is governed by interactions — the same pH raises removal for one pollutant–material pair and lowers it for another — which is what motivates flexible learners over univariate screening in the first place, and it cautions against reading the univariate signs mechanistically.

Spearman correlations among the numeric descriptors (Fig. S1) show the expected blocks: the Abraham solvation parameters correlate strongly with molecular weight and with each other, and the composition-averaged elemental descriptors correlate with the metal identities they are derived from. Variance inflation factors make the redundancy explicit — 22 of 26 numeric descriptors exceed the usual threshold of 10, with a median VIF near 7×10^4 (Table S6) — because descriptors are constant within a study, so the study-cluster structure induces near-perfect collinearity. This redundancy motivates the feature-selection experiment of Section 3.4 and, more fundamentally, explains why record-level splits leak: any two records of the same study are near-duplicates in feature space. Missingness is confined to the synthesis and texture blocks (Table 1): calcination conditions are unreported for two thirds of the records, and pH_{pzc} and pore volume for roughly a quarter, which is why imputation quality matters and why texture-based mechanistic claims from compiled data warrant caution.

3.2. Within-system interpolation (RQ1)

Table 3 answers (RQ1) by comparing the seven models and the naive baselines under task (T1), reported as mean $\pm$ SD over 20 seeded splits.

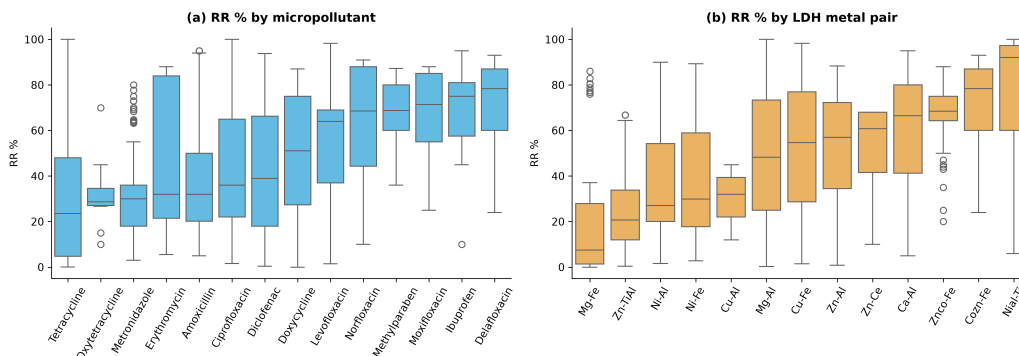


Figure 2: Removal-ratio distributions by pollutant and by metal pair

324 The tree ensembles lead: XGBoost and HistGB reach test R^2 of 0.84–0.85
 325 with RMSE near 11 percentage points, followed by Cubist and RF, while LR
 326 trails far behind. This is because RR% responds nonlinearly to the operating
 327 conditions, with thresholds in pH and dose that additive linear terms cannot
 328 represent. The baselines put these numbers in perspective. The per-study
 329 mean, which uses no descriptors at all, already attains R^2 of 0.43 (MAE
 330 16.7): nearly half of the variance a record-level split appears to explain is
 331 available from system identity alone, and LR barely improves on it. The
 332 ensembles' further gain to R^2 of 0.85 therefore reflects genuine modeling of
 333 the operating conditions within known systems. The in-sample R^2 of the
 334 ensembles approaches 0.99, and the gap to the test score is the expected
 335 overfitting margin of high-capacity learners rather than a protocol artifact;
 336 the cross-validation and test scores agree closely. Figure 3 shows the parity
 337 and residual plots of the best model on the reference split. It is worth noting
 338 that these scores are already 2–3 RMSE points more conservative than a
 339 protocol without duplicate removal on the same data, since deduplication
 340 alone eliminates records whose test values could be looked up verbatim from

341 training.

342 3.3. Cross-study generalization (RQ2)

343 Figure 4 answers (RQ2) with the leave-one-study-out audit. The picture
344 changes sharply: the median per-study MAE of the best models rises to 14–
345 16 percentage points, two to three times their interpolation error, and the
346 per-study R^2 is positive for only 4 of the 32 studies large enough to evaluate
347 it. This is because the models exploit descriptor combinations that identify
348 a material–pollutant system rather than transferable structure–activity rela-
349 tions; when all records of a system are withheld, that shortcut disappears.
350 The baseline comparison adds an important nuance. Descriptor-free lookups
351 transfer even worse: the global training-mean reaches a median per-study
352 MAE of 24 points and the per-pollutant mean of 25, so the models do reduce
353 the naive error by roughly 40 percent, and Cubist beats the per-pollutant
354 lookup in 41 of the 59 studies. The models therefore track the level differences
355 between systems while failing to resolve the condition response within an un-
356 seen system, which is exactly why the per-study R^2 (measured against each
357 study’s own variance) stays negative even as the MAE gain over the baselines
358 is large. Screening applications get a coarse ranking signal from these mod-
359 els, not engineering precision. The per-study distribution in Fig. 4b is wide
360 rather than uniformly poor: for Cubist, 20 of the 59 studies are predicted
361 with errors comparable to the interpolation regime (MAE below 6 points),
362 while 9 studies exceed 30 points. Notably, the model ranking also reorders:
363 Cubist, mid-table under interpolation, attains the lowest cross-study median,
364 which could be explained by its rule-plus-linear structure extrapolating more
365 gracefully than deep tree ensembles.

366 **Which systems transfer.** Figure 6 breaks the per-study errors down
367 by chemistry, and the pattern follows literature density. Ciprofloxacin stud-
368 ies, which dominate the compilation (32 of 59 studies), are predicted with
369 a median MAE of 5 points, and Zn–Al systems (28 studies) with 5 points,
370 because each held-out study of these families leaves many close relatives in
371 training. Sparsely covered chemistries fare far worse: tetracycline, diclofenac,
372 and levofloxacin studies show median errors above 20 points, and Ni–Fe sys-
373 tems above 30. Error also grows with the held-out study’s own scope: per-
374 study MAE correlates with both the number of records and the within-study
375 spread of RR% ($r \approx 0.45$ for each), meaning studies that explore wide con-
376 dition ranges are precisely the ones an outside model cannot resolve. The
377 learning curve in Fig. 5b is flat between 5 and 40 training studies, suggesting
378 that cross-study error is limited by the diversity of compiled systems rather
379 than by record count. This supports our central argument that random-split
380 scores, including those in Table 3, must not be read as screening skill: the
381 same models, the same data, and the same preprocessing yield near-zero skill
382 when the test set contains genuinely new systems.

383 3.4. Feature selection under both tasks

384 Table 4 evaluates the three nested descriptor sets of Section 2.4 under
385 both tasks with fixed hyperparameters. Selection is close to neutral in both
386 directions. Interpolation accuracy is indistinguishable across the sets (R^2 of
387 0.85 ± 0.03 for HistGB in all three; XGBoost and Cubist behave the same,
388 Table S8), so ten well-chosen descriptors carry essentially all the within-
389 system signal of the full thirty. This is because the removable descriptors are
390 redundant with the retained ones — the collinear Abraham and composition-

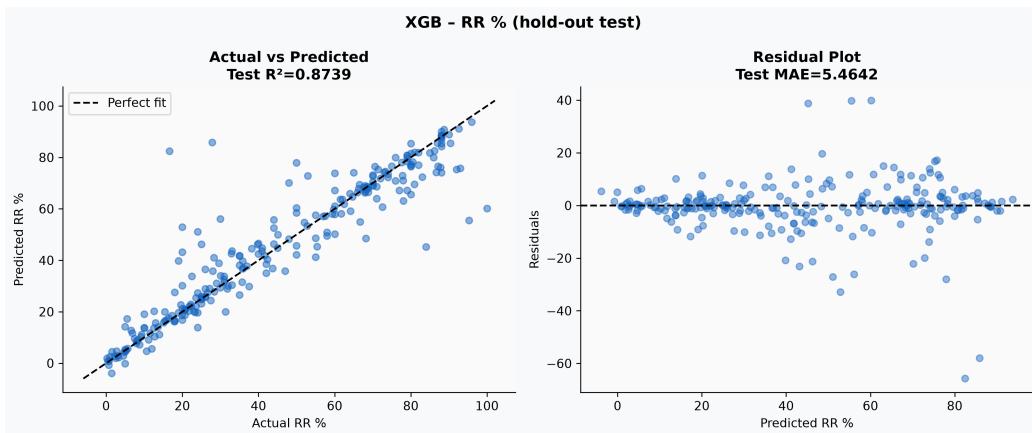


Figure 3: Parity and residual plots of the best model under interpolation

391 averaged block flagged by the VIF analysis — rather than independently in-
 392 formative. In the other direction, no reduction rescues cross-study transfer:
 393 the LOSO median moves only from 16.4 to 16.2 points, consistent with the
 394 flat learning curve — the transfer bottleneck is study diversity, not dimen-
 395 sionality. The practical reading favors parsimony: the top-10 set consists of
 396 the five operating conditions, four texture properties, and $\log K_{ow}$, all rou-
 397 tinely reported quantities, so a deployable model does not require the full
 398 descriptor battery.

399 3.5. Uncertainty and applicability domain (RQ3)

400 Figure 7 answers (RQ3). Under interpolation, the split-conformal 90%
 401 intervals of Eq. (9) achieve empirical coverage of 93% with a half-width of
 402 19 percentage points, so predictions for studied systems can be certified at
 403 a practically useful precision. Under the new-study task, coverage is main-
 404 tained (99%) only because the study-grouped calibration inflates the half-
 405 width to 68 points, an interval spanning most of the 0–100 RR% range and

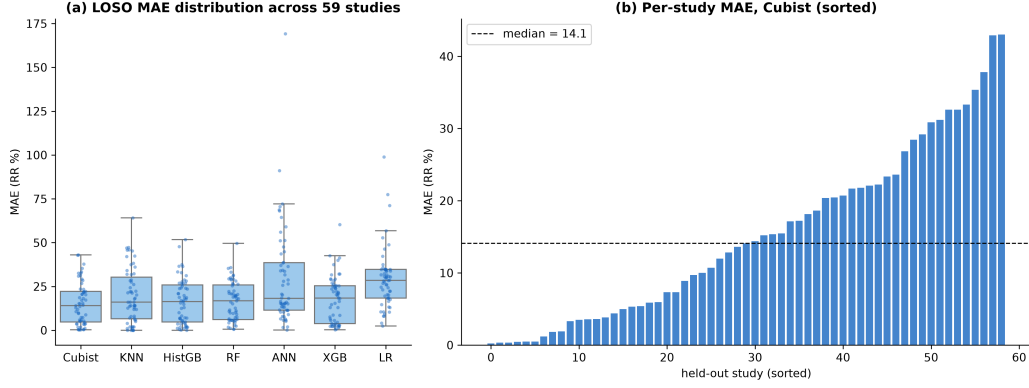


Figure 4: Leave-one-study-out audit across 59 studies

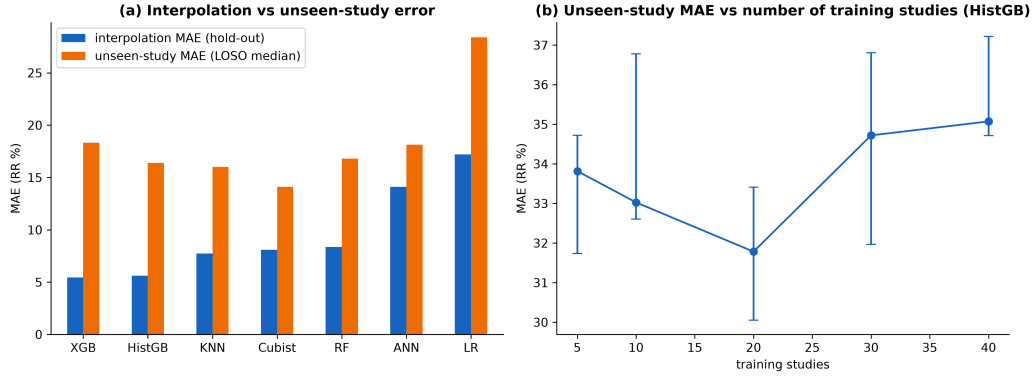


Figure 5: Interpolation versus unseen-study error and learning curve

406 therefore carrying little decision value. This is because residual magnitudes
 407 differ systematically between systems, so a single global quantile must cover
 408 the worst of them. The distance-based applicability-domain flag does not
 409 compensate: the mean distance to the five nearest training records carries es-
 410 sentially no information about realized error in either task ($r \leq 0.06$, Fig. 7b),
 411 although such distance heuristics are widely used as trust flags. A key find-
 412 ing is therefore that the uncertainty tools inherit the transfer problem itself:
 413 interpolation predictions can be certified, whereas no construction we tested

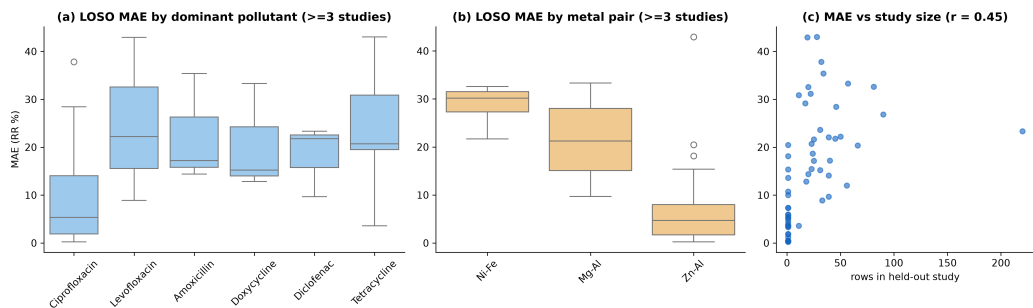


Figure 6: Cross-study error by pollutant, metal pair, and study size

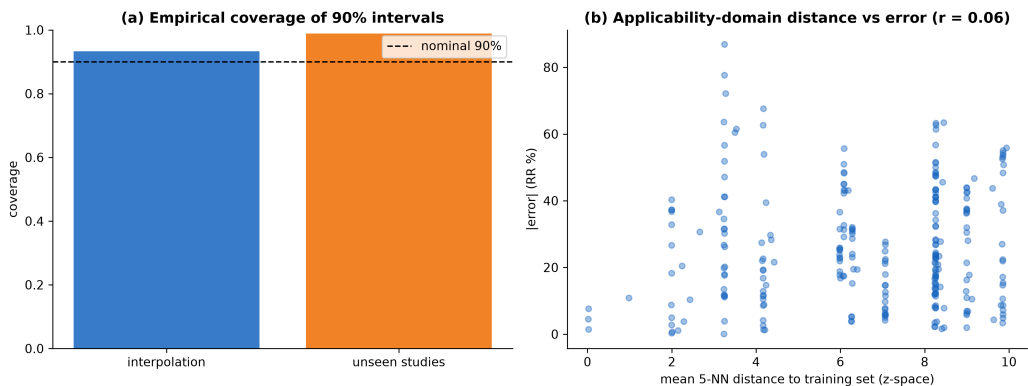


Figure 7: Conformal coverage and applicability-domain flag

414 produces informative guarantees for unseen systems, and coverage claims
 415 must be validated empirically for the deployment task.

416 3.6. Feature effects and mechanistic reading (RQ4)

417 Figure 8 and Table 5 answer (RQ4). Adding the chemistry-informed fea-
 418 tures leaves both tasks essentially unchanged (interpolation MAE 5.6 versus
 419 5.8 points; cross-study median 16.4 versus 16.0), indicating that the models
 420 already reconstruct these combinations internally from pH, pH_{pzc} , and dose.
 421 Their value here is interpretive rather than predictive.

422 **Operating conditions dominate.** Permutation-importance ranks on
 423 the hold-out set (Table 5) agree across six of the seven models on the leading
 424 factors: adsorbent dose ranks first, followed by contact time, initial concen-
 425 tration, and pH, with the first material property (BET) at median rank six
 426 and the first pollutant descriptor ($\log K_{ow}$) at rank nine; only the additive LR
 427 deviates. The SHAP summary (Fig. 8, top) shows the same hierarchy from
 428 a second attribution method, and aggregating mean $|\text{SHAP}|$ by descriptor
 429 group makes it quantitative: operating conditions account for 54% of the to-
 430 tal attribution mass, material texture for 18%, composition and synthesis for
 431 14%, and pollutant descriptors for 14% (Table S7). The partial-dependence
 432 profiles of the three leading features (Fig. S2) show textbook adsorption be-
 433 havior: removal rises steeply with dose up to roughly 2–5 g/L and then
 434 saturates, rises within the first hours of contact and plateaus at equilibrium,
 435 and declines with initial concentration beyond roughly 200 mg/L as fixed
 436 capacity is exhausted. Within known systems, the models therefore learn
 437 genuine condition–response physics. The same ranking, however, explains
 438 the transfer failure of Section 3.3 from a second direction: material and pol-
 439 lutant descriptors, the only information available about an unseen system,
 440 contribute comparatively little to the fitted models.

441 **Mechanistic reading.** The SHAP dependence of $\text{pH} - \text{pH}_{\text{pzc}}$ (Fig. 8a)
 442 is not smoothly monotone but threshold-like: contributions stay within a
 443 few points while the surface remains positively charged, and a sharp penalty
 444 of up to nine points appears once solution pH exceeds pH_{pzc} by more than
 445 about three units. Coloring this dependence by $\log K_{ow}$ (Fig. S3) shows the
 446 penalty concentrated among hydrophilic compounds, consistent with elec-

447 trostatic repulsion of anionic species from a negatively charged surface while
448 hydrophobic partitioning buffers the loss for lipophilic ones. Log K_{ow} itself
449 is the strongest chemical attribution (Fig. 8b), monotone from roughly -5
450 points for the most hydrophilic compounds to $+15$ for the most lipophilic.
451 BET surface area (Fig. 8c) is a cautionary counterexample: the model assigns
452 positive contributions to records with low measured BET even though the
453 raw association runs the other way (median RR% of 44 below $20 \text{ m}^2/\text{g}$ versus
454 50 above), so this conditional attribution reflects the correlation structure of
455 the compilation rather than a surface-area mechanism, and we do not inter-
456 pret it mechanistically. Attributions for the composition-averaged elemental
457 descriptors are small, consistent with the finding that current composition
458 descriptors carry little transferable information between studies. [MECHA-
459 NISM PARAGRAPH FLAGGED FOR DOMAIN REVIEW BY THE AU-
460 THORS.]

461 3.7. Computation time

462 Training cost is not a differentiator at this data scale. On the reference
463 split, a single fit takes below one second for every model except XGBoost (7–
464 10 s across repeated measurements with the tuned 495-tree configuration),
465 and prediction takes below 0.1 s throughout (Table S3). The full leave-one-
466 study-out audit, 59 refits of all seven models, completes within minutes on a
467 workstation, so the honest grouped protocol carries no meaningful computa-
468 tional premium over the random-split protocol it replaces. Hyperparameter
469 search with Optuna dominates the overall budget and is performed once, not
470 per evaluation task.

471 *3.8. Environmental implications and outlook*

472 For practice, the results delimit two use cases sharply. Water-treatment
473 engineers optimizing operating conditions for an LDH-pollutant system that
474 has been measured before can rely on literature-trained models with a mean
475 absolute error near 6 percentage points of RR%, tightened further by confor-
476 mal intervals. Materials researchers seeking to screen new LDH formulations
477 or new pollutants face a harder trade: the models do carry transferable sig-
478 nal, reducing naive baseline errors by roughly 40 percent, but per-study skill
479 remains near zero, so predictions for unmeasured systems support coarse
480 prioritization at best and should be treated as hypotheses for experiment
481 rather than estimates. Reported R^2 values above 0.9 in prior compilations
482 (TODO, 0000f,a) are consistent with our record-level numbers and, by the
483 mechanism quantified here, likely overstate screening ability in the same way;
484 re-evaluation of published models under grouped protocols would clarify how
485 general this pattern is.

486 We see three concerns that temper the conclusions. First, the compila-
487 tion covers 59 studies, so the cross-study estimates themselves carry sampling
488 uncertainty; the per-study distribution, not a single mean, is the honest sum-
489 mary. The second concern is reporting heterogeneity between laboratories
490 (differing protocols for pH_{pzc} or BET), which acts as irreducible noise under
491 (T2) and may partly explain the transfer gap. The third concern is that our
492 descriptors, like those of prior compilations, describe the bulk material; in-
493 terlayer anion identity and surface morphology are rarely reported and may
494 carry the transferable signal that current models lack. In future work, richer
495 composition-structure descriptors, study-level random effects, and few-shot

496 calibration on a handful of measurements of a new system are the most
497 promising routes toward genuine screening capability.

498 4. Conclusions

499 We evaluated seven machine learning models for micropollutant adsorp-
500 tion on layered double hydroxides under a leakage-aware protocol that sepa-
501 rates within-system interpolation from cross-study generalization, using 1,342
502 experiments compiled from 59 studies. The models predict removal ratio for
503 studied systems at new operating conditions with test R^2 of 0.85 ± 0.03
504 and RMSE near 11 percentage points across 20 repeated splits, but the same
505 models show near-zero per-study skill when entire studies are held out, and a
506 leave-one-study-out audit locates the failure in a wide per-study error distri-
507 bution rather than a uniform degradation. Against descriptor-free baselines
508 the picture is more nuanced: the models reduce the naive cross-study error
509 by roughly 40 percent, so they do carry transferable signal, yet not enough
510 to certify predictions for an unmeasured system. Split-conformal intervals
511 certify interpolation predictions at ± 19 percentage points, whereas for un-
512 seen studies neither conformal calibration nor a distance-based applicability
513 score yields informative guarantees. Chemistry-informed features such as
514 $\text{pH} - \text{pH}_{\text{pzc}}$ leave the accuracy essentially unchanged but anchor the learned
515 behavior in electrostatic and hydrophobic mechanisms. In future work, we
516 plan to extend the compilation with interlayer-anion and morphology de-
517 scriptors, model study-level effects explicitly, and test few-shot calibration
518 so that a small number of measurements of a new system can recover the
519 accuracy currently reserved for studied ones.

520 **CRedit authorship contribution statement**

521 [TO BE COMPLETED BY THE AUTHORS.]

522 **Declaration of competing interest**

523 The authors declare that they have no known competing financial inter-
524 ests or personal relationships that could have appeared to influence the work
525 reported in this paper.

526 **Data availability**

527 The curated dataset, the full modeling pipeline (`ldh_pipeline.py`), and
528 the notebooks reproducing every table and figure are available at [REPOSI-
529 TORY URL TO BE ADDED].

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Table 1: Summary statistics of the numeric descriptors

Descriptor	Median	Range	Missing (%)
<i>Micropollutant</i>			
$\log K_{ow}$	−0.02	−2.67–4.51	0
MW (g/mol)	331	152–734	0
$E / S / A / B / V$	1.81 / 2.10 / 0.70 / 1.70 / 2.30	–	0
TPSA (Å ²)	76.2	37.3–194.0	0
<i>Composition and synthesis</i>			
M(II)/M(III) ratio	3.0	0.5–8.0	0
Avg. atomic weight (g/mol)	48.1	24.8–73.7	0
Avg. d-electrons	4.25	0–8.89	0
Avg. Pauling electronegativity	1.59	1.30–1.89	0
Hydrothermal temp. (°C)	70	25–160	13.3
Hydrothermal time (h)	24	12–40	13.3
Calcination temp. (°C)	400	200–550	65.9
Calcination time (h)	4	1–12	65.9
<i>Material texture</i>			
d_{003} (nm)	0.77	0.25–1.13	2.5
BET (m ² /g)	82.0	0.4–225.7	7.9
PV (cm ³ /g)	0.10	0–1.46	25.3
APD (nm)	8.4	0.2–64.2	13.9
pH _{pzc}	7.5	5.5–11.6	24.6
<i>Operating conditions</i>			
pH	6.7	2.0–12.0	0
Temperature (°C)	25	10–55	0
Contact time (h)	1.5	0.02–192	0
Dose (g/L)	0.9	0.01–50	0
C_i (mg/L)	40	1–1688	0

Table 2: Spearman rank correlation of numeric descriptors with RR% (twelve largest by magnitude)

Descriptor	ρ	Descriptor	ρ
PV (cm ³ /g)	0.43	TPSA	-0.19
Dose (g/L)	0.33	S	-0.19
E	-0.30	C_i (mg/L)	0.18
APD (nm)	0.28	d_{003} (nm)	0.17
pH	0.27	B	-0.16
Hydrothermal time (h)	0.23	Calcination time (h)	0.16

Table 3: Model and baseline performance under within-system interpolation (T1), mean $\pm$ SD over 20 seeded 80/20 splits [VALUES FROM repeated_interp_summary.csv]

Model	Test R^2	Test MAE	Test RMSE
XGBoost	0.84 ± 0.03	6.3 ± 0.6	11.1 ± 1.0
HistGB	0.85 ± 0.03	6.3 ± 0.5	10.8 ± 1.0
KNN	0.73 ± 0.06	7.8 ± 0.8	14.4 ± 1.5
Cubist	0.79 ± 0.03	8.0 ± 0.6	12.7 ± 1.0
RF	0.78 ± 0.03	8.6 ± 0.5	13.0 ± 0.9
ANN	0.54 ± 0.05	14.4 ± 0.7	19.1 ± 0.9
LR	0.40 ± 0.04	17.4 ± 0.5	21.7 ± 0.7
Per-study mean	0.43 ± 0.05	16.7 ± 0.8	21.2 ± 0.9
Per-pollutant mean	0.18 ± 0.03	21.4 ± 0.6	25.4 ± 0.6
Global mean	-0.01 ± 0.01	24.9 ± 0.7	28.1 ± 0.6

Table 4: Feature-set comparison under both tasks (HistGB; interpolation mean $\pm$ SD over 20 splits, LOSO median over 59 studies)

Feature set	p	Interp. R^2	Interp. MAE	LOSO median MAE
Full	30	0.85 ± 0.03	6.3 ± 0.5	16.4
Correlation-screened	21	0.85 ± 0.03	6.4 ± 0.5	16.2
Top-10 permutation	10	0.85 ± 0.03	6.4 ± 0.5	16.2

Table 5: Permutation-importance ranks of the leading descriptors across the seven models (hold-out test set) [VALUES FROM perm_ranks_all_models.csv]

Descriptor	LR	KNN	ANN	Cubist	RF	XGB	HistGB
AD (dose)	27	1	29	1	1	1	1
Time	19	3	1	2	2	2	2
C_i	22	4	14	3	3	4	3
pH	25	2	4	5	7	5	4
BET	24	5	2	7	6	6	6
pH _{pzc}	26	7	19	9	4	7	5
d_{003}	8	6	8	8	12	8	10
Avg. electronegativity	15	13	7	15	9	9	8

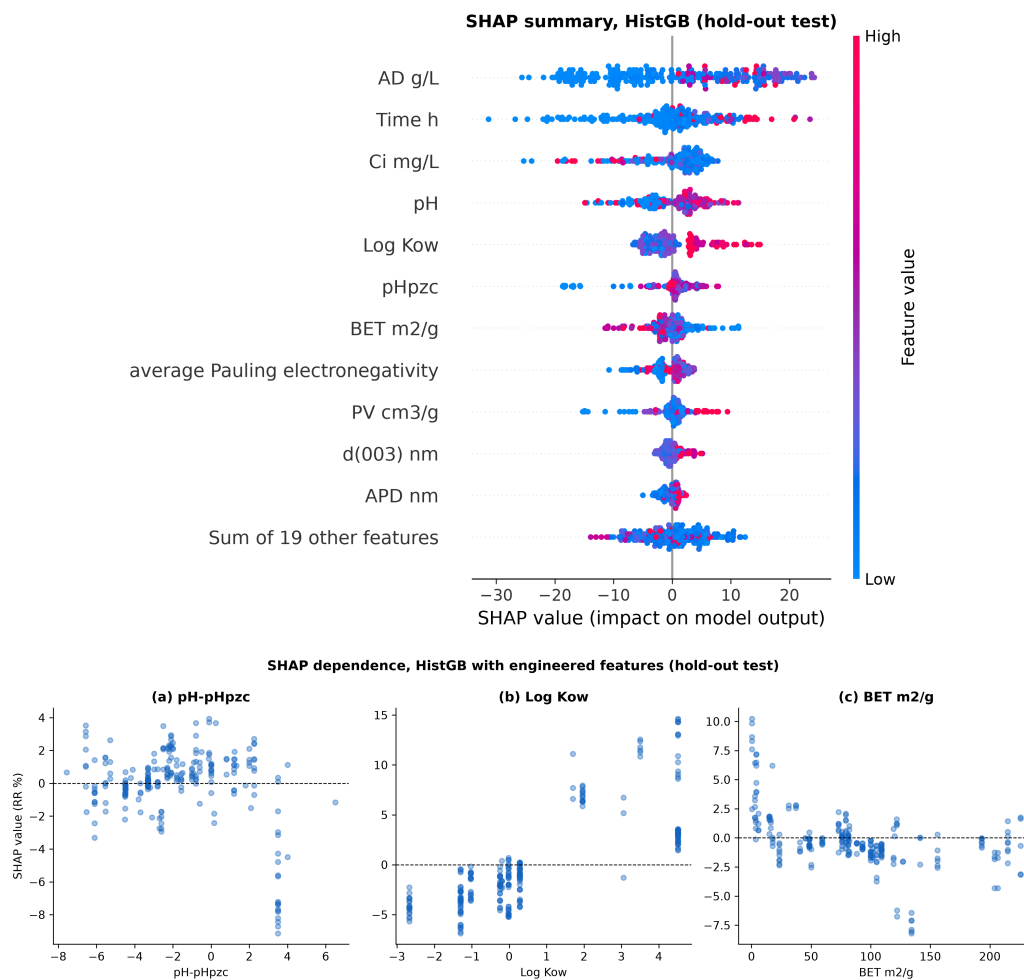


Figure 8: SHAP summary (top) and dependence of mechanism-linked features (bottom)